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Susceptibility of the Instrumentation to Conducted EMI Injected Through the Ground Plane

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***Abstract*—This article presents the effects of electromagnetic interferences in common instrumentation, like multimeters and oscilloscopes. This investigation is of paramount importance when the interferences can directly reach the instrumentation used to verify the susceptibility of the circuits under test. In particular, the referred case of study is that of electromagnetic interferences injected through the ground plane into the output terminals of the analog integrated circuits. The measurements show that the susceptibility of the instrumentation depends on the amplitude, the coupling, and the frequency, but generally it remains low and mostly negligible.**

***Index Terms*—Analog integrated circuits, electromagnetic immunity, injection points, instrumentation, multimeters, oscilloscopes.**

I. Introduction

The vulnerability of electronic circuits to the electromagnetic interferences (EMI) is a critical issue, since electronic is nowadays ubiquitous [1]. The number of possible sources, victims, and paths for the interferences is, therefore, increasing continuously. Recently, the EMI susceptibility has been further investigated in nontraditional scenarios: for example, by considering the EMI susceptibility of also digital circuits and not only of the analog ones [2], [3], [4], [5], or by experimentally evaluating the effect of the interferences from all the possible points of injections [6], [7], [8], [9], [10]. In this latter case, interesting and unexpected results have been obtained by capacitive coupling the interferences to the output pin of the devices under test. An overview of the susceptibility measurements obtained for some commercial amplifiers is visible in the plots of Fig. 1. In this test [2], four commercial amplifiers have been exposed to EMI in different conditions of amplitude (from 500 mVp up to 2 Vp) and frequency (from 1 MHz up to 1 GHz). On the *y*-axis it is plotted the EMI-induced offset of the amplifiers' output voltage. The results show a strong susceptibility, dependent on the amplitude (an expected outcome) and on the frequency (less predictable), with detrimental effects on all the circuits driven by the amplifiers.

A similar investigation has been carried out on voltage reference circuits, as well [8]. The measurement results, in the case of interferences capacitively coupled from the ground plane, are plotted in Fig. 2.

It is, therefore, evident the high susceptibility of commercial ICs not only in the case of direct injection into the input or power supply pins, but also in the case of interferences coupling from the ground plane into the output pin. This particular scenario of injection could rise objections because the effect of the interferences is measured in the same point of the injection. This issue is illustrated in Fig. 3 in an immediate and intuitive way. In the first case [see Fig. 3(a)], the interferences reach only the DUT through its input pin and the instrumentation measures the output signal of the DUT; in the second case [see Fig. 3(b)], the injection point is the same of the measurement point and the interferences could theoretically influence the instrumentation behavior.

This correspondence is therefore focused on experimental evidence of the EMI immunity of common instrumentation, and, in particular,

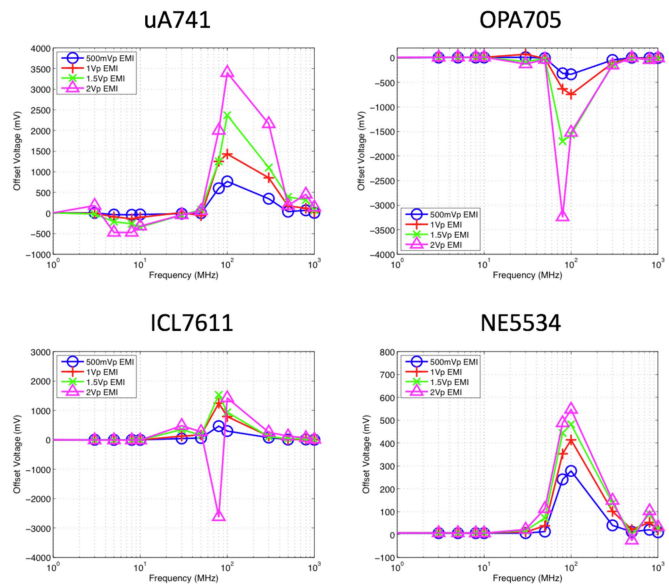

Fig. 1. EMI-induced offset in commercial opamps [2].

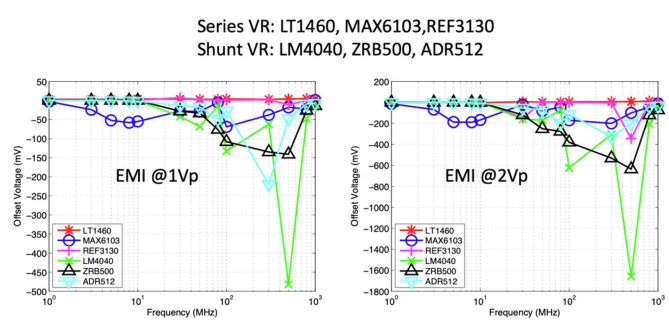

Fig. 2. EMI-induced offset in series and shunt commercial voltage references [8].

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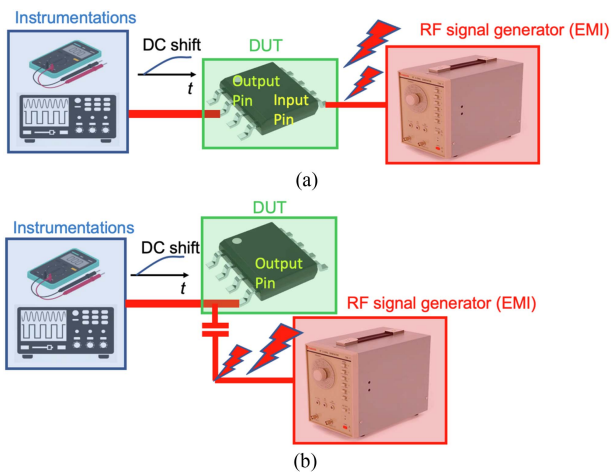

Fig. 3. EMI paths and effects on the instrumentation. (a) In the case of classical injection points. (b) In the case of novel injection points.

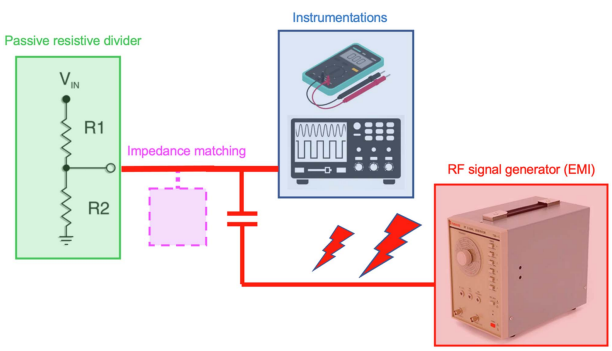

Fig. 4. EMI test to verify the instrumentation immunity.

it aims to confirm the results obtained in [7], [8], [9], and [10]. In the case of active components indeed, the measurements clearly show the effect of the interferences, as in Figs. 1 and 2. However, the focus of this article is to demonstrate that the results are not affected by the sensibility of the instrumentation itself. Indeed, if the instrumentation shows a good immunity level, the measurements reported in [7], [8], [9], and [10] are verified; otherwise, the latter would be meaningless.

II. Immunity Test of the Instrumentation

The proposed EMI immunity test is based on a simple resistive divider replacing the DUT (as in Fig. 4), and it can be run as a first step when the instrumentation is used in EMC issues. The divider is biased at $V_{\text{in}} = 5$ V by an E3630A Hewlett-Packard power supply and it provides nominal (i.e., without EMI) 2.58 V: two SMD resistors of 6.8 kΩ are chosen to limit the current below milliampere. To investigate the EMI effect on the instrumentation, the interferences are sinusoidal signals, with 0DC mean value, amplitude of 2 Vpp and frequency from 250 kHz up to 4 GHz, as in [1]. The interferences are provided by the RF signal generator Hewlett-Packard E4422B. EMI are injected into the output path through an SMD capacitor of 4.7 pF, a realistic value of the parasitic capacitance between a trace and the PCB ground plane [11], [12], [13]. An impedance matching network can be also added to further test the vulnerability of the instrumentation. It is evident that the tested instrumentation (the multimeter and the oscilloscope) shows a susceptibility to the EMI negligible if compared to that of the active devices and ICs (as in Fig. 5). Thus, the instrumentation cannot

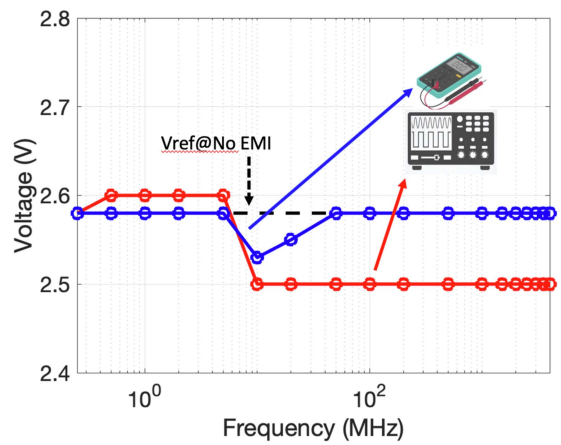

Fig. 5. Results of the immunity test of the instrumentation.

influence the EMI-induced offset in the analog ICs reported in [7], [8], [9], and [10] and the strong susceptibility is mainly due to the parasitic capacitances of the PCB traces that can deeply modify the frequency response of the ICs.

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